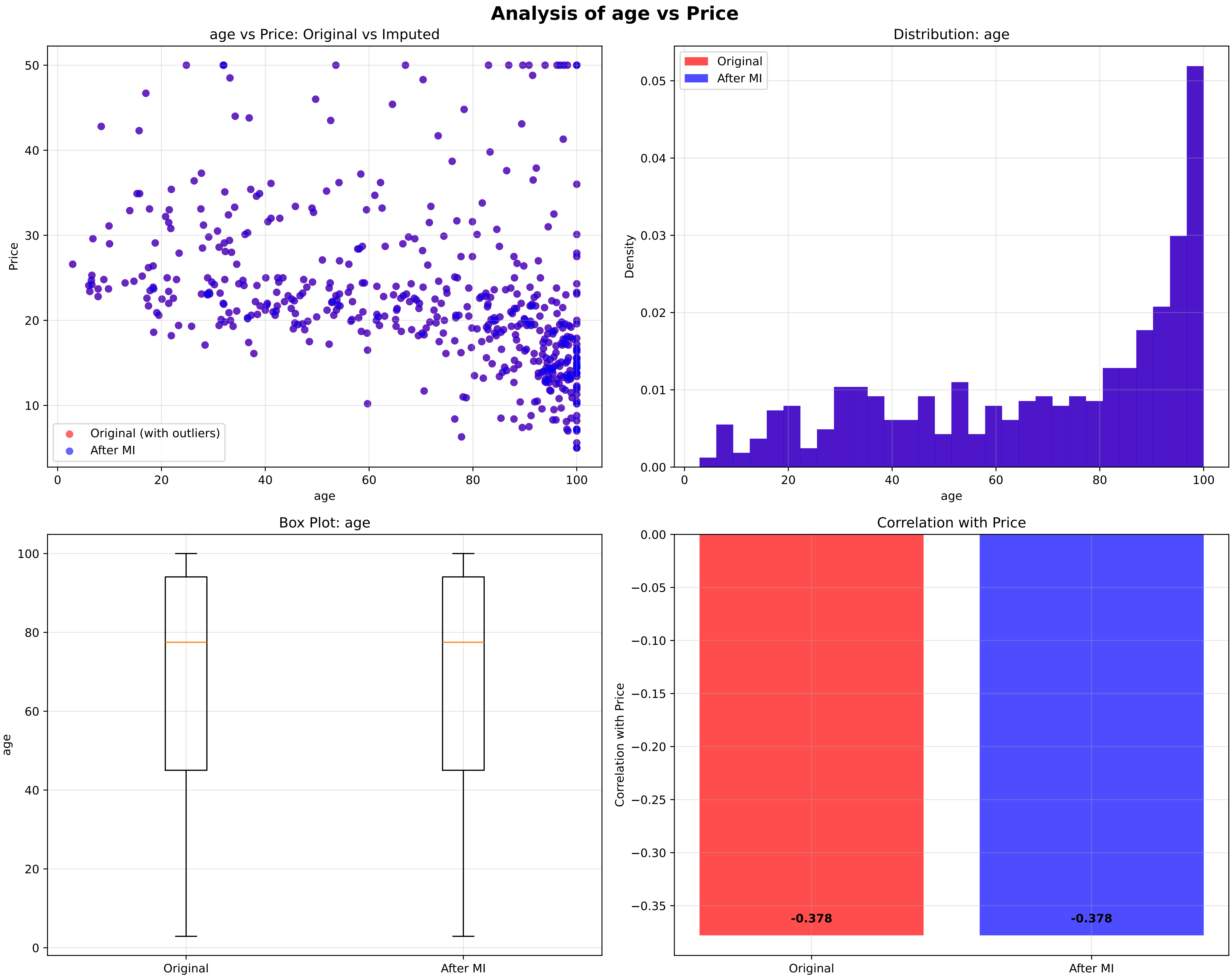

Analysis of age vs Price
age vs Price: Original vs Imputed
Price
age
Original (with outliers)
After MI
Distribution: age
Density
age
Original
After MI
Box Plot: age
age
Original
After MI
Correlation with Price
Correlation with Price
-0.378
-0.378
Original
After MI

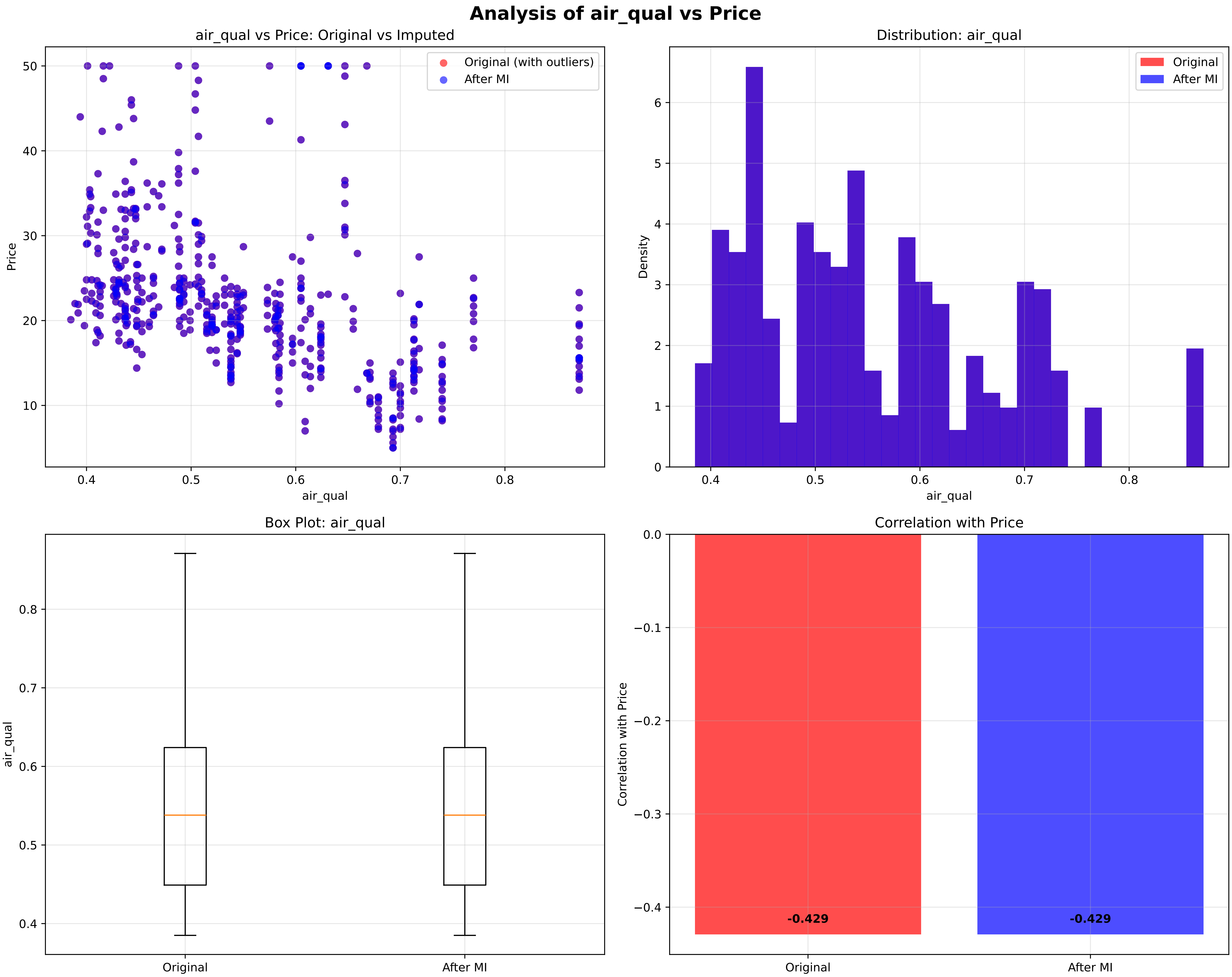
STATISTICAL SUMMARY: age vs Price
==

ORIGINAL DATA:
- Mean age: 68.575
- Std age: 28.149
- Median age: 77.500
- Correlation with Price: -0.378
- Outliers detected: 0

AFTER MULTIPLE IMPUTATION:
- Mean age: 68.575
- Std age: 28.149
- Median age: 77.500
- Correlation with Price: -0.378

REGRESSION ANALYSIS:
- Original R^2: 0.143
- Imputed R^2: 0.143
- Change in correlation: +0.000

INTERPRETATION:
Newer properties (lower age) usually have higher values. Minimal correlation change after outlier handling (+0.000).


Analysis of air_qual vs Price
air_qual vs Price: Original vs Imputed
Original (with outliers)
After MI
Price
air_qual
Distribution: air_qual
Original
After MI
Density
air_qual
Box Plot: air_qual
air_qual
Original
After MI
Correlation with Price
Correlation with Price
-0.429
-0.429
Original
After MI

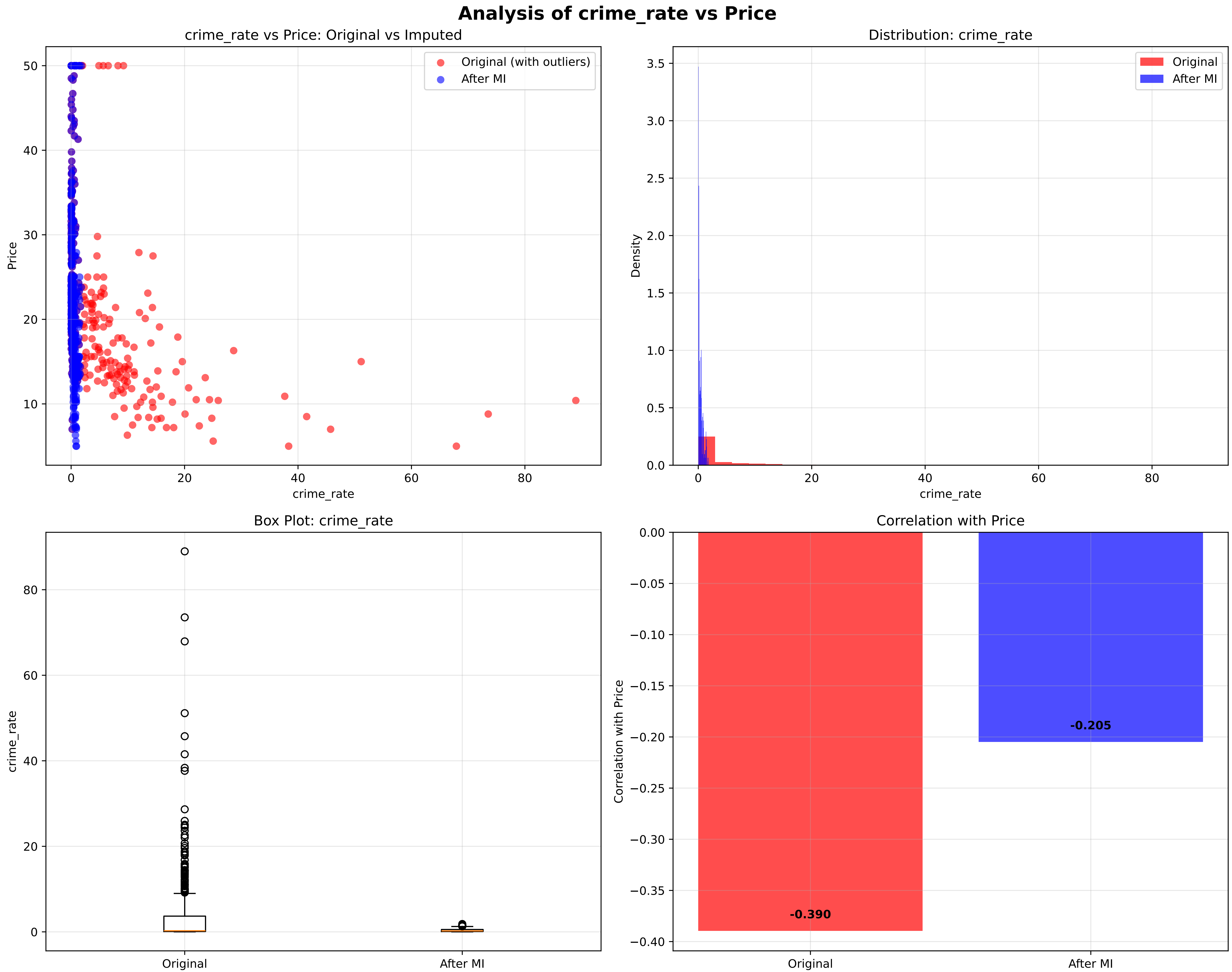
STATISTICAL SUMMARY: air_qual vs Price
==

ORIGINAL DATA:
- Mean air_qual: 0.555
- Std air_qual: 0.116
- Median air_qual: 0.538
- Correlation with Price: -0.429
- Outliers detected: 0

AFTER MULTIPLE IMPUTATION:
- Mean air_qual: 0.555
- Std air_qual: 0.116
- Median air_qual: 0.538
- Correlation with Price: -0.429

REGRESSION ANALYSIS:
- Original R^2: 0.184
- Imputed R^2: 0.184
- Change in correlation: +0.000

INTERPRETATION:
Better air quality is positively correlated with higher prices. Minimal correlation change after outlier handling (+0.000



Analysis of crime_rate vs Price
crime_rate vs Price: Original vs Imputed
Original (with outliers)
After MI
Price
crime_rate
Distribution: crime_rate
Original
After MI
Density
crime_rate
Box Plot: crime_rate
crime_rate
Original
After MI
Correlation with Price
Correlation with Price
-0.390
-0.205
Original
After MI

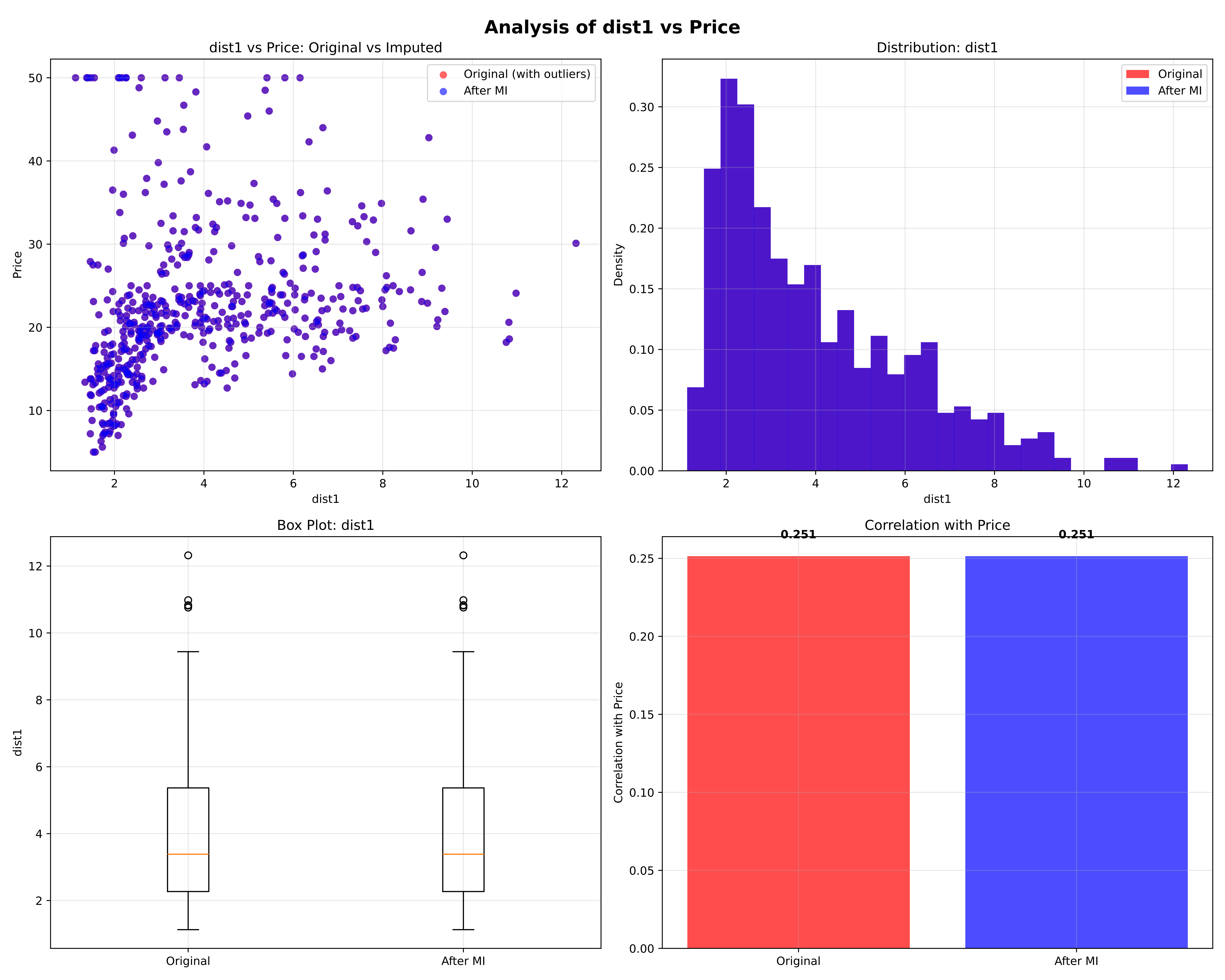
STATISTICAL SUMMARY: crime_rate vs Price
==

ORIGINAL DATA:
- Mean crime_rate: 3.614
- Std crime_rate: 8.602
- Median crime_rate: 0.257
- Correlation with Price: -0.390
- Outliers detected: 150

AFTER MULTIPLE IMPUTATION:
- Mean crime_rate: 0.389
- Std crime_rate: 0.395
- Median crime_rate: 0.229
- Correlation with Price: -0.205

REGRESSION ANALYSIS:
- Original R^2: 0.152
- Imputed R^2: 0.042
- Change in correlation: +0.185

INTERPRETATION:
Higher crime rates typically negatively affect property prices. Significant correlation change after outlier handling (+0



Analysis of dist1 vs Price
dist1 vs Price: Original vs Imputed
Original (with outliers)
After MI
Price
dist1
Distribution: dist1
Original
After MI
Density
dist1
Box Plot: dist1
dist1
Original
After MI
Correlation with Price
0.251
0.251
Correlation with Price
Original
After MI

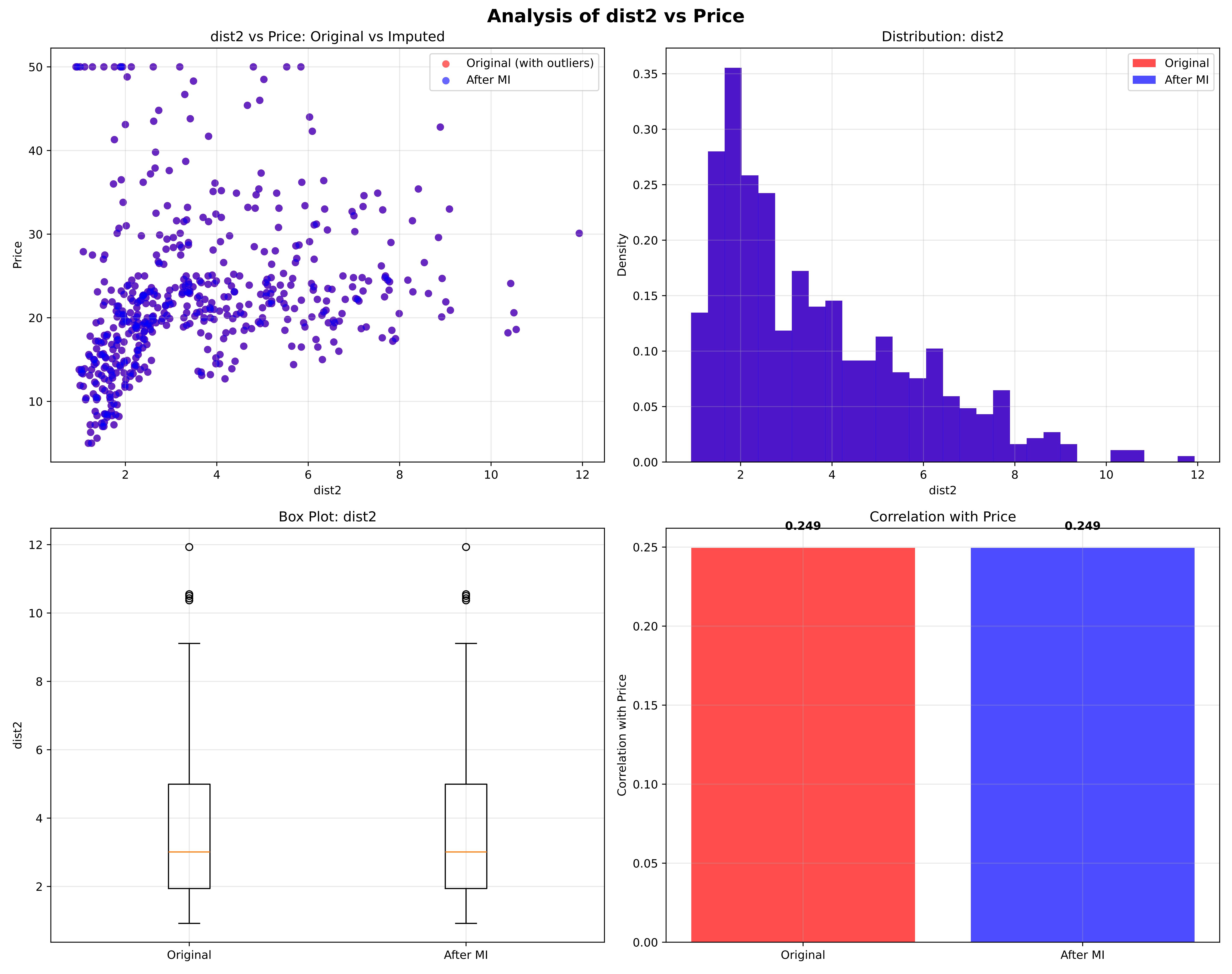
```
STATISTICAL SUMMARY: dist1 vs Price
==================================================

ORIGINAL DATA:
- Mean dist1: 3.972
- Std dist1: 2.109
- Median dist1: 3.385
- Correlation with Price: 0.251
- Outliers detected: 0

AFTER MULTIPLE IMPUTATION:
- Mean dist1: 3.972
- Std dist1: 2.109
- Median dist1: 3.385
- Correlation with Price: 0.251

REGRESSION ANALYSIS:
- Original R²: 0.063
- Imputed R²: 0.063
- Change in correlation: +0.000

INTERPRETATION:
Distance to feature 1 impact varies by context. Minimal correlation change after outlier handling (+0.000).
```


Analysis of dist2 vs Price
dist2 vs Price: Original vs Imputed
Original (with outliers)
After MI
Price
dist2
Distribution: dist2
Original
After MI
Density
dist2
Box Plot: dist2
dist2
Original
After MI
Correlation with Price
0.249
0.249
Correlation with Price
Original
After MI

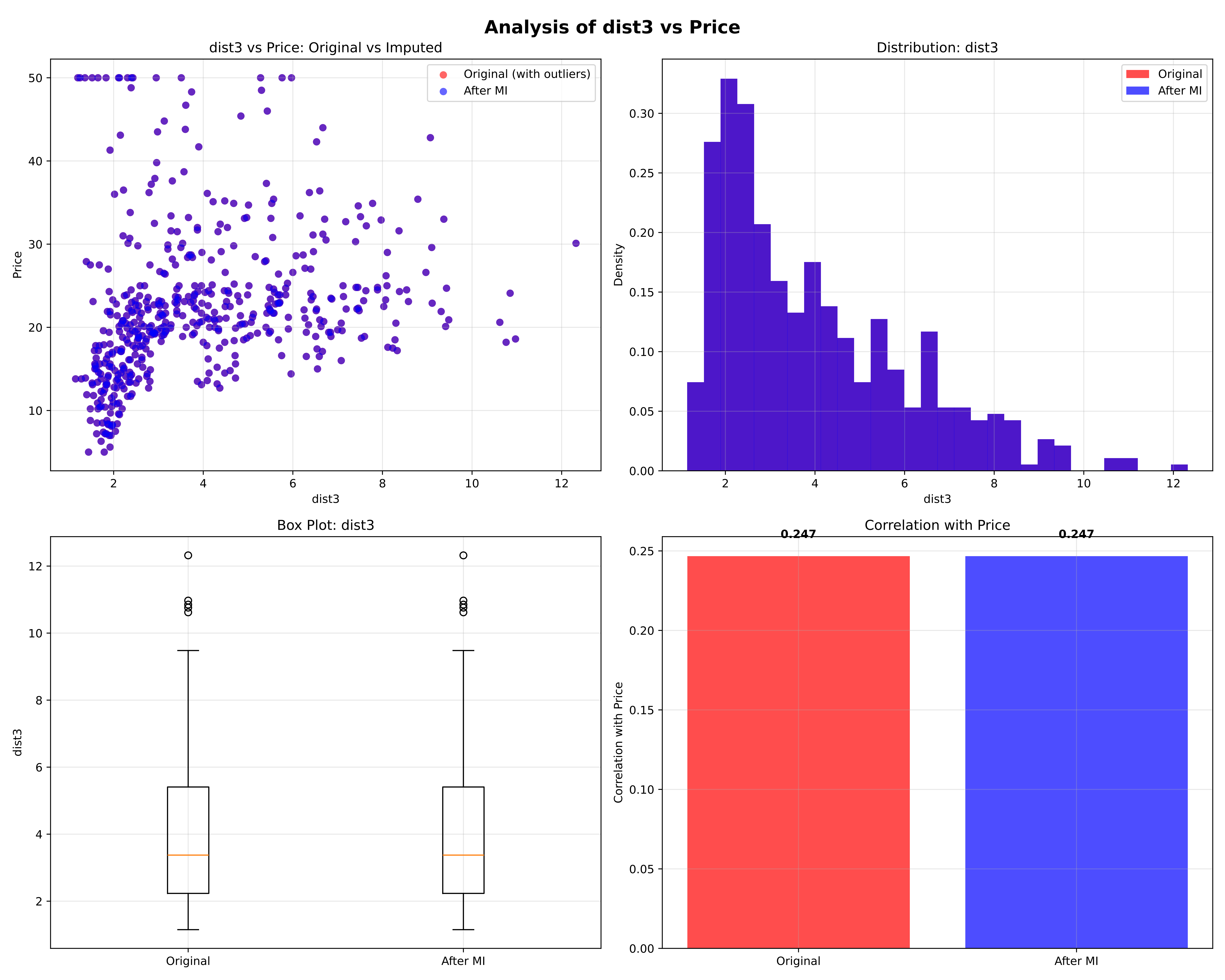
```
STATISTICAL SUMMARY: dist2 vs Price
==================================================

ORIGINAL DATA:
- Mean dist2: 3.629
- Std dist2: 2.109
- Median dist2: 3.010
- Correlation with Price: 0.249
- Outliers detected: 0

AFTER MULTIPLE IMPUTATION:
- Mean dist2: 3.629
- Std dist2: 2.109
- Median dist2: 3.010
- Correlation with Price: 0.249

REGRESSION ANALYSIS:
- Original R²: 0.062
- Imputed R²: 0.062
- Change in correlation: +0.000

INTERPRETATION:
Variable impact on price depends on context. Minimal correlation change after outlier handling (+0.000).
```


Analysis of dist3 vs Price
dist3 vs Price: Original vs Imputed
Original (with outliers)
After MI
Price
dist3
Distribution: dist3
Original
After MI
Density
dist3
Box Plot: dist3
dist3
Original
After MI
Correlation with Price
0.247
0.247
Correlation with Price
Original
After MI

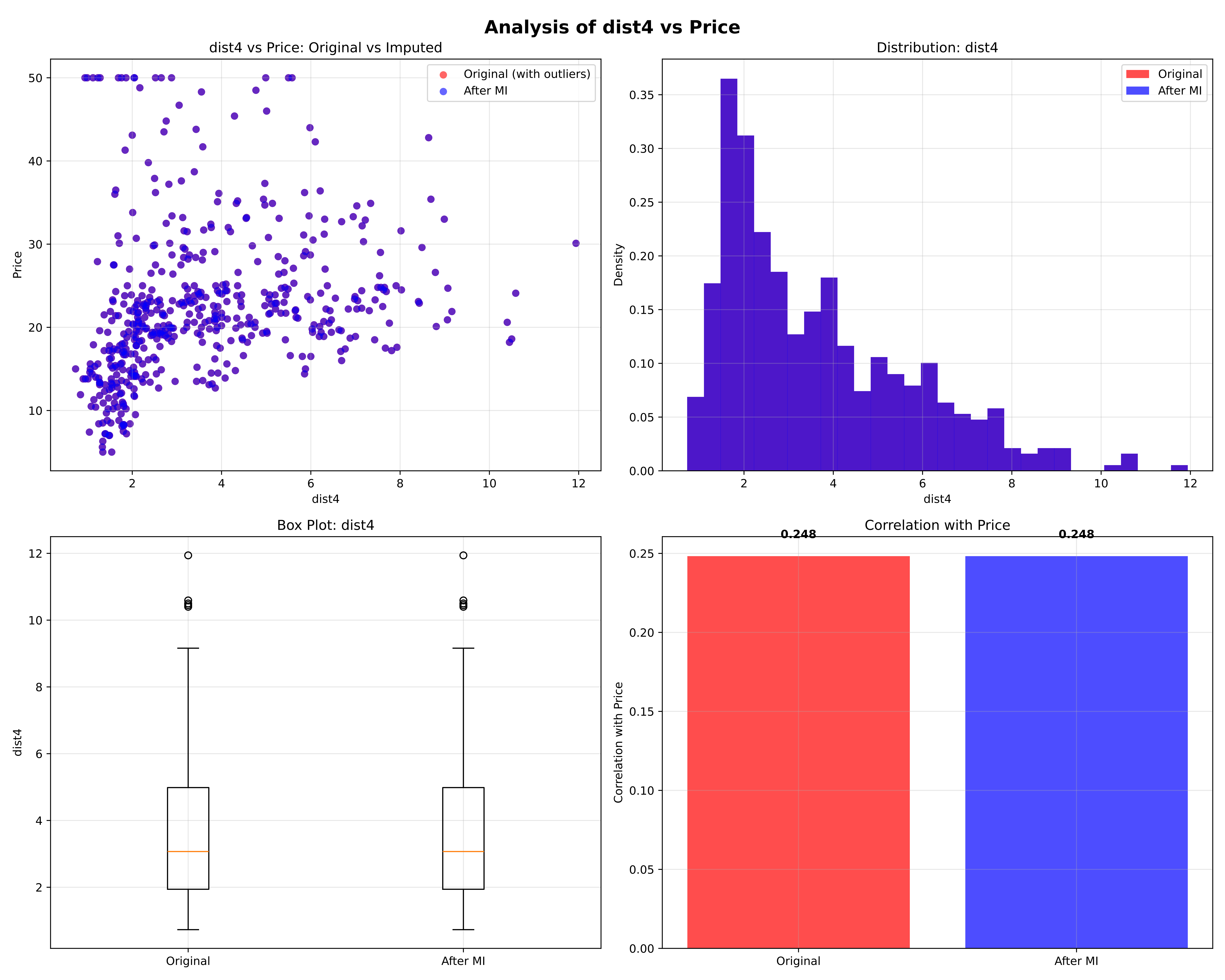
STATISTICAL SUMMARY: dist3 vs Price
==

ORIGINAL DATA:
- Mean dist3: 3.961
- Std dist3: 2.120
- Median dist3: 3.375
- Correlation with Price: 0.247
- Outliers detected: 0

AFTER MULTIPLE IMPUTATION:
- Mean dist3: 3.961
- Std dist3: 2.120
- Median dist3: 3.375
- Correlation with Price: 0.247

REGRESSION ANALYSIS:
- Original R^2: 0.061
- Imputed R^2: 0.061
- Change in correlation: +0.000

INTERPRETATION:
Variable impact on price depends on context. Minimal correlation change after outlier handling (+0.000).



Analysis of dist4 vs Price
dist4 vs Price: Original vs Imputed
Original (with outliers)
After MI
Price
dist4
Distribution: dist4
Original
After MI
Density
dist4
Box Plot: dist4
dist4
Original
After MI
Correlation with Price
0.248
0.248
Correlation with Price
Original
After MI

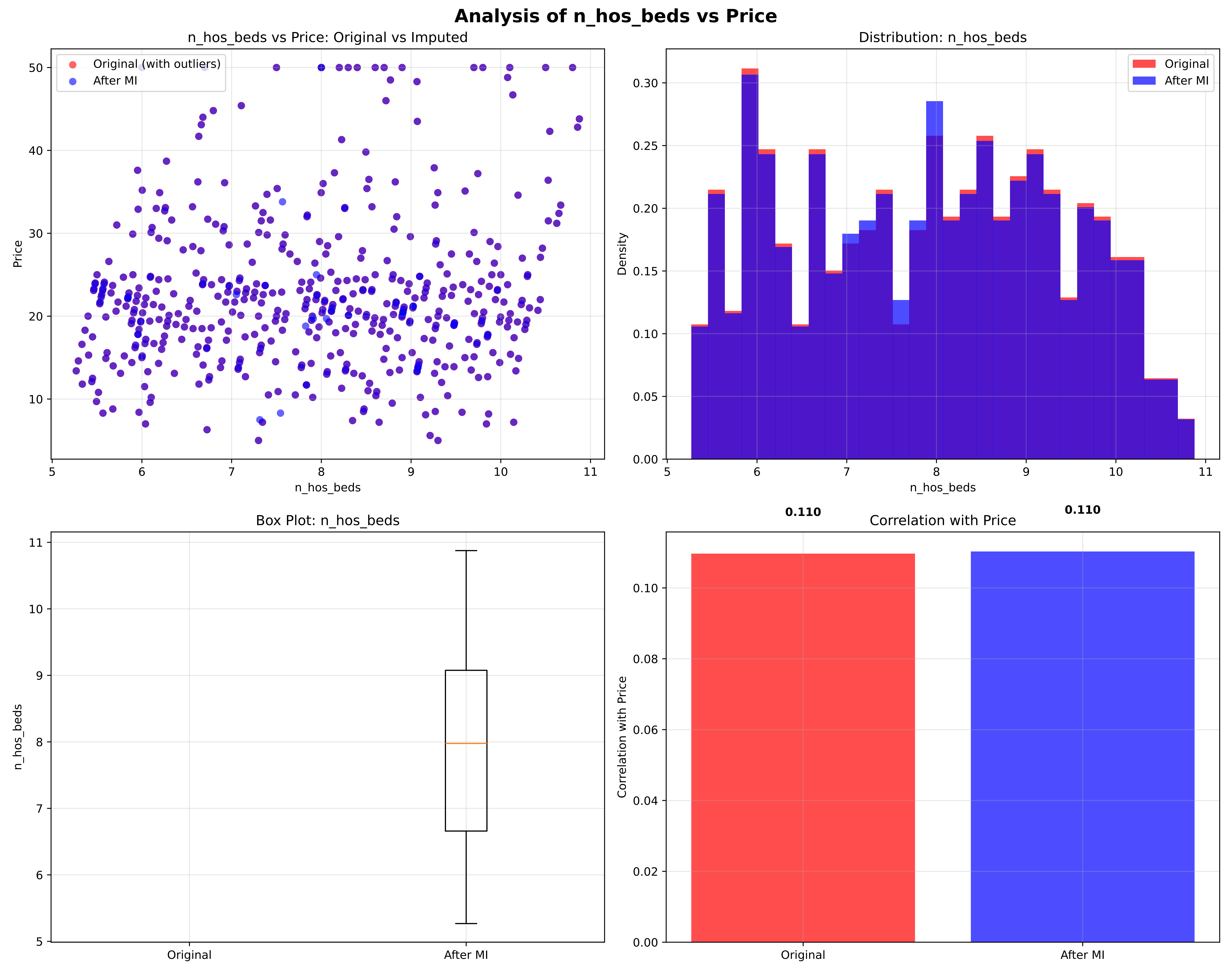
STATISTICAL SUMMARY: dist4 vs Price
==

ORIGINAL DATA:
- Mean dist4: 3.619
- Std dist4: 2.099
- Median dist4: 3.070
- Correlation with Price: 0.248
- Outliers detected: 0

AFTER MULTIPLE IMPUTATION:
- Mean dist4: 3.619
- Std dist4: 2.099
- Median dist4: 3.070
- Correlation with Price: 0.248

REGRESSION ANALYSIS:
- Original R^2: 0.062
- Imputed R^2: 0.062
- Change in correlation: +0.000

INTERPRETATION:
Variable impact on price depends on context. Minimal correlation change after outlier handling (+0.000).



Analysis of n_hos_beds vs Price
n_hos_beds vs Price: Original vs Imputed
Original (with outliers)
After MI
Price
n_hos_beds
Distribution: n_hos_beds
Original
After MI
Density
n_hos_beds
Box Plot: n_hos_beds
n_hos_beds
Original
After MI
Correlation with Price
0.110
0.110
Correlation with Price
Original
After MI

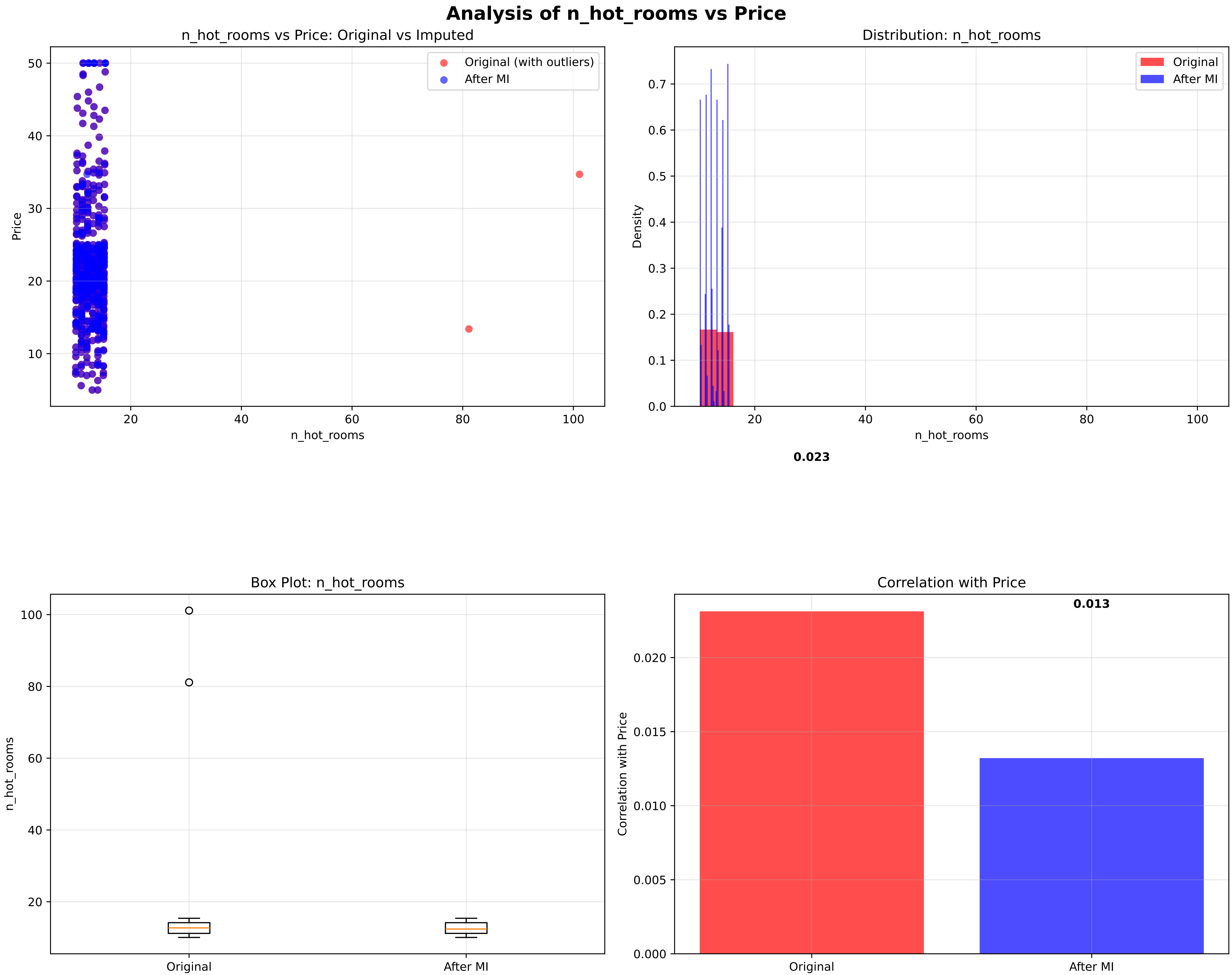
STATISTICAL SUMMARY: n_hos_beds vs Price
==

ORIGINAL DATA:
- Mean n_hos_beds: 7.900
- Std n_hos_beds: 1.477
- Median n_hos_beds: 7.999
- Correlation with Price: 0.110
- Outliers detected: 0

AFTER MULTIPLE IMPUTATION:
- Mean n_hos_beds: 7.896
- Std n_hos_beds: 1.466
- Median n_hos_beds: 7.977
- Correlation with Price: 0.110

REGRESSION ANALYSIS:
- Original R^2: nan
- Imputed R^2: 0.012
- Change in correlation: +0.001

INTERPRETATION:
More hospital beds may indicate better healthcare access. Minimal correlation change after outlier handling (+0.001).



Analysis of n_hot_rooms vs Price
n_hot_rooms vs Price: Original vs Imputed
Original (with outliers)
After MI
Price
n_hot_rooms
Distribution: n_hot_rooms
Original
After MI
Density
n_hot_rooms
0.023
Box Plot: n_hot_rooms
n_hot_rooms
Original
After MI
Correlation with Price
0.013
Correlation with Price
Original
After MI

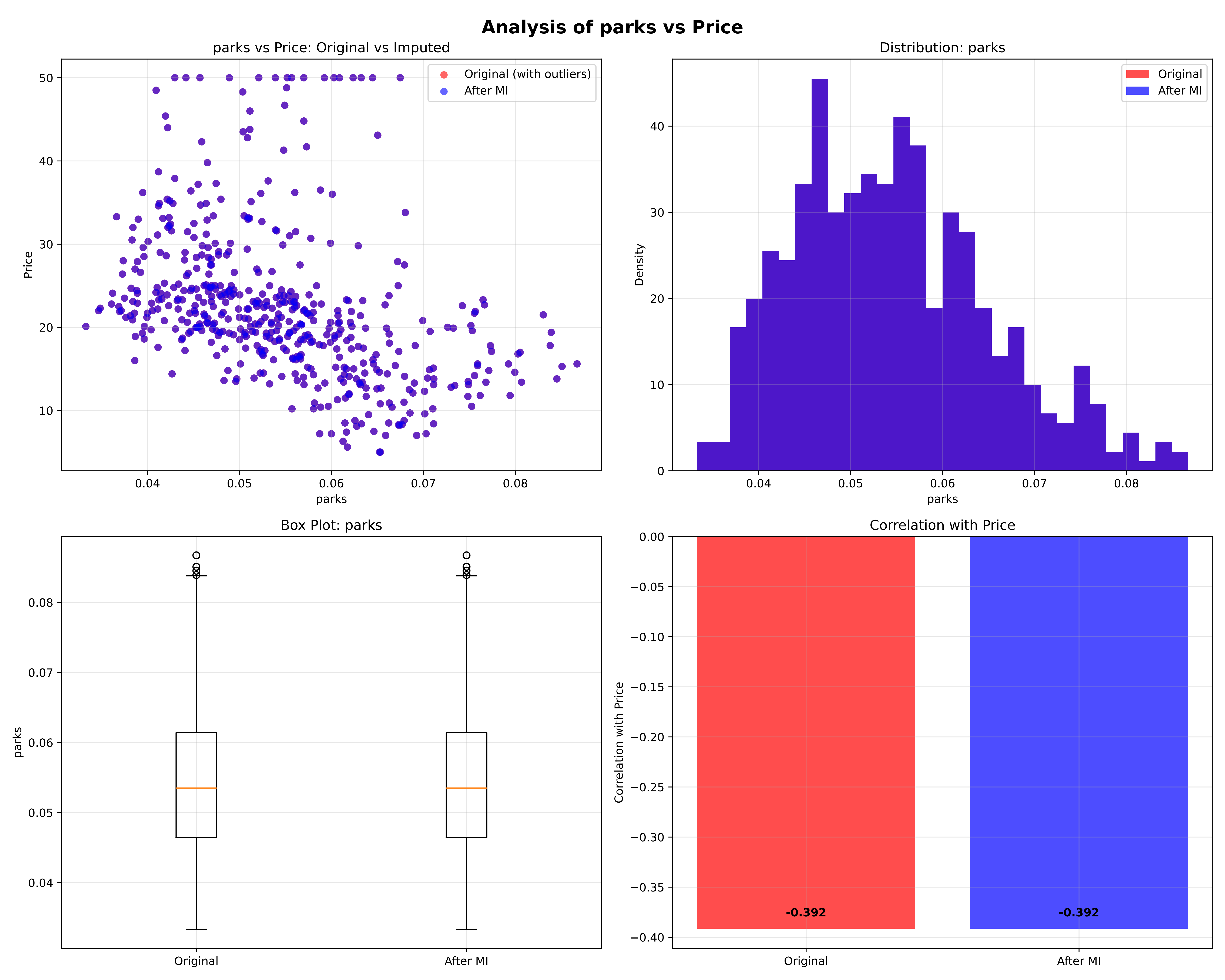
STATISTICAL SUMMARY: n_hot_rooms vs Price
==

ORIGINAL DATA:
- Mean n_hot_rooms: 13.042
- Std n_hot_rooms: 5.239
- Median n_hot_rooms: 12.720
- Correlation with Price: 0.023
- Outliers detected: 2

AFTER MULTIPLE IMPUTATION:
- Mean n_hot_rooms: 12.731
- Std n_hot_rooms: 1.677
- Median n_hot_rooms: 12.400
- Correlation with Price: 0.013

REGRESSION ANALYSIS:
- Original R^2: 0.001
- Imputed R^2: 0.000
- Change in correlation: -0.010

INTERPRETATION:
More hotel rooms may indicate tourist area effects. Minimal correlation change after outlier handling (-0.010).



Analysis of parks vs Price
parks vs Price: Original vs Imputed
Original (with outliers)
After MI
Price
parks
Distribution: parks
Original
After MI
Density
parks
Box Plot: parks
parks
Original
After MI
Correlation with Price
Correlation with Price
-0.392
-0.392
Original
After MI

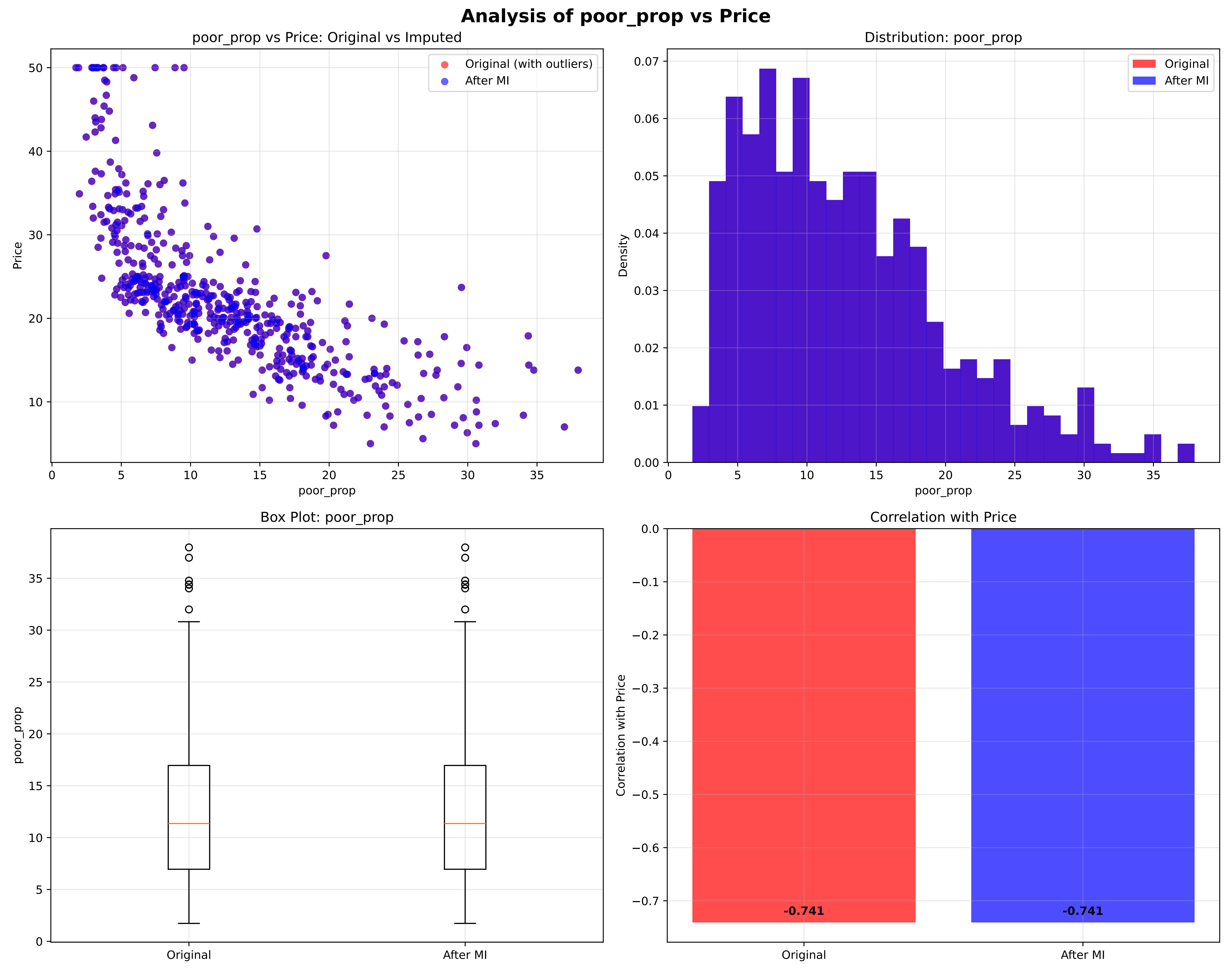
STATISTICAL SUMMARY: parks vs Price
==

ORIGINAL DATA:
- Mean parks: 0.054
- Std parks: 0.011
- Median parks: 0.054
- Correlation with Price: -0.392
- Outliers detected: 0

AFTER MULTIPLE IMPUTATION:
- Mean parks: 0.054
- Std parks: 0.011
- Median parks: 0.054
- Correlation with Price: -0.392

REGRESSION ANALYSIS:
- Original R^2: 0.153
- Imputed R^2: 0.153
- Change in correlation: +0.000

INTERPRETATION:
More parks generally increase property values. Minimal correlation change after outlier handling (+0.000).


Analysis of poor_prop vs Price
poor_prop vs Price: Original vs Imputed
Original (with outliers)
After MI
Price
poor_prop
Distribution: poor_prop
Original
After MI
Density
poor_prop
Box Plot: poor_prop
poor_prop
Original
After MI
Correlation with Price
Correlation with Price
-0.741
-0.741
Original
After MI

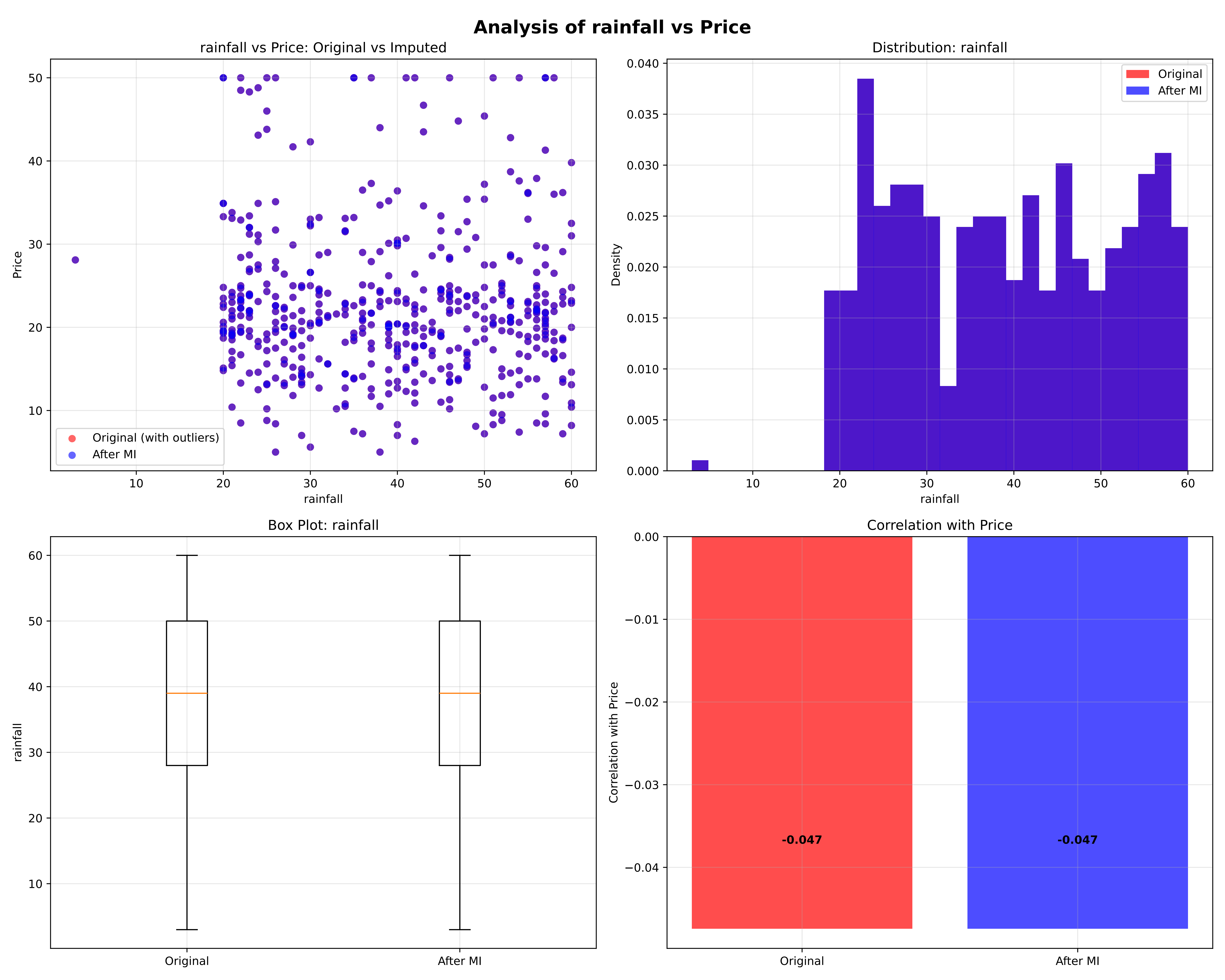
```
STATISTICAL SUMMARY: poor_prop vs Price
==================================================

ORIGINAL DATA:
- Mean poor_prop: 12.653
- Std poor_prop: 7.141
- Median poor_prop: 11.360
- Correlation with Price: -0.741
- Outliers detected: 0

AFTER MULTIPLE IMPUTATION:
- Mean poor_prop: 12.653
- Std poor_prop: 7.141
- Median poor_prop: 11.360
- Correlation with Price: -0.741

REGRESSION ANALYSIS:
- Original R²: 0.549
- Imputed R²: 0.549
- Change in correlation: +0.000

INTERPRETATION:
Higher poverty proportion typically decreases prices. Minimal correlation change after outlier handling (+0.000).
```



Analysis of rainfall vs Price
rainfall vs Price: Original vs Imputed
Price
rainfall
Original (with outliers)
After MI
Distribution: rainfall
Density
Original
After MI
Box Plot: rainfall
Original
After MI
Correlation with Price
Correlation with Price
-0.047
-0.047

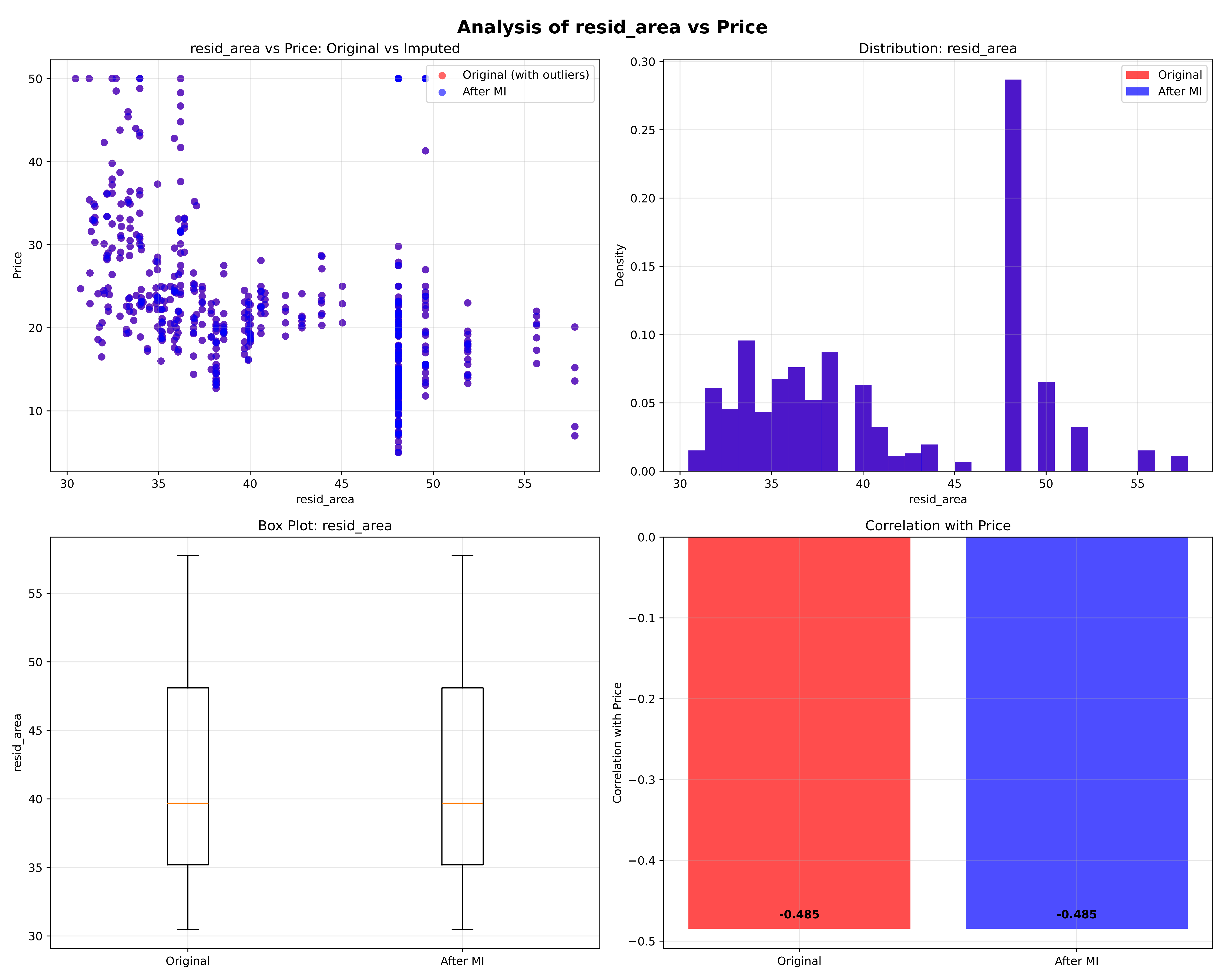
STATISTICAL SUMMARY: rainfall vs Price
==

ORIGINAL DATA:
- Mean rainfall: 39.182
- Std rainfall: 12.514
- Median rainfall: 39.000
- Correlation with Price: -0.047
- Outliers detected: 0

AFTER MULTIPLE IMPUTATION:
- Mean rainfall: 39.182
- Std rainfall: 12.514
- Median rainfall: 39.000
- Correlation with Price: -0.047

REGRESSION ANALYSIS:
- Original R^2: 0.002
- Imputed R^2: 0.002
- Change in correlation: +0.000

INTERPRETATION:
Rainfall impact depends on regional preferences. Minimal correlation change after outlier handling (+0.000).


Analysis of resid_area vs Price
resid_area vs Price: Original vs Imputed
Original (with outliers)
After MI
Price
resid_area
Distribution: resid_area
Original
After MI
Density
resid_area
Box Plot: resid_area
resid_area
Original
After MI
Correlation with Price
Correlation with Price
-0.485
-0.485
Original
After MI

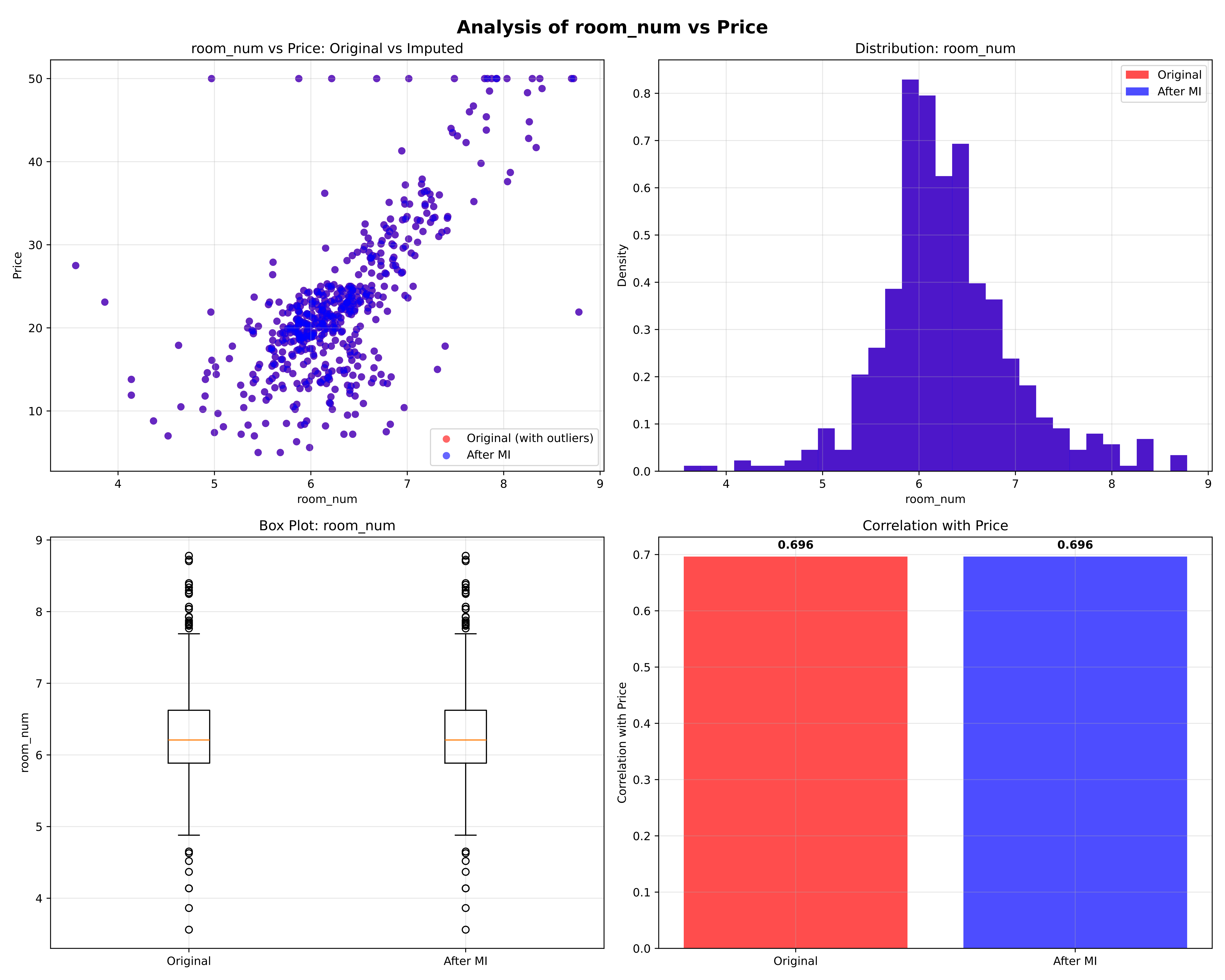
STATISTICAL SUMMARY: resid_area vs Price
==

ORIGINAL DATA:
- Mean resid_area: 41.137
- Std resid_area: 6.860
- Median resid_area: 39.690
- Correlation with Price: -0.485
- Outliers detected: 0

AFTER MULTIPLE IMPUTATION:
- Mean resid_area: 41.137
- Std resid_area: 6.860
- Median resid_area: 39.690
- Correlation with Price: -0.485

REGRESSION ANALYSIS:
- Original R^2: 0.235
- Imputed R^2: 0.235
- Change in correlation: +0.000

INTERPRETATION:
Larger residential areas generally increase property value. Minimal correlation change after outlier handling (+0.000).


Analysis of room_num vs Price
room_num vs Price: Original vs Imputed
Price
room_num
Original (with outliers)
After MI
Distribution: room_num
Density
Original
After MI
Box Plot: room_num
Original
After MI
Correlation with Price
Correlation with Price
0.696
0.696
Original
After MI

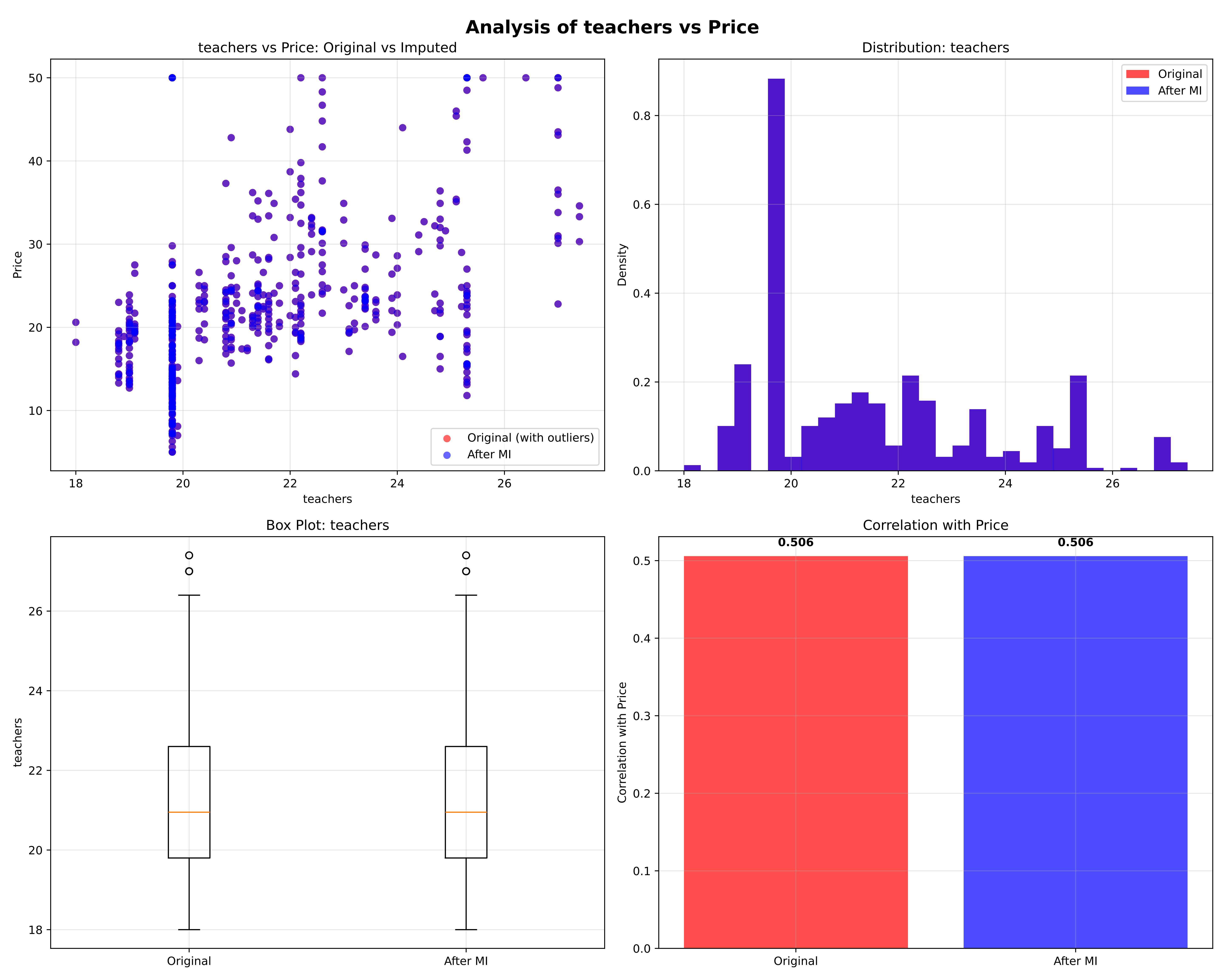
```
STATISTICAL SUMMARY: room_num vs Price
==================================================

ORIGINAL DATA:
- Mean room_num: 6.285
- Std room_num: 0.703
- Median room_num: 6.208
- Correlation with Price: 0.696
- Outliers detected: 0

AFTER MULTIPLE IMPUTATION:
- Mean room_num: 6.285
- Std room_num: 0.703
- Median room_num: 6.208
- Correlation with Price: 0.696

REGRESSION ANALYSIS:
- Original R²: 0.485
- Imputed R²: 0.485
- Change in correlation: +0.000

INTERPRETATION:
More rooms typically increase property value. Minimal correlation change after outlier handling (+0.000).
```



Analysis of teachers vs Price
teachers vs Price: Original vs Imputed
Price
teachers
Original (with outliers)
After MI
Distribution: teachers
Density
teachers
Original
After MI
Box Plot: teachers
teachers
Original
After MI
Correlation with Price
Correlation with Price
0.506
0.506
Original
After MI

STATISTICAL SUMMARY: teachers vs Price
==

ORIGINAL DATA:
- Mean teachers: 21.544
- Std teachers: 2.165
- Median teachers: 20.950
- Correlation with Price: 0.506
- Outliers detected: 0

AFTER MULTIPLE IMPUTATION:
- Mean teachers: 21.544
- Std teachers: 2.165
- Median teachers: 20.950
- Correlation with Price: 0.506

REGRESSION ANALYSIS:
- Original R^2: 0.256
- Imputed R^2: 0.256
- Change in correlation: +0.000

INTERPRETATION:
More teachers may indicate better education, increasing prices. Minimal correlation change after outlier handling (+0.000